

EE 611: VLSI Technology

Course Code	EE 611
Course Name	VLSI Technology
L-T-P-C	3-0-0-3
Prerequisites	EE 160 Applied Electronics, EE 208P Digital System Design Practicum, Semiconductor devices or Instructors consent
Intended for	4th year B. Tech. (EE) and MS, Ph.D.
Distribution	Elective
Approval	5th Senate (Oct 2013)

Course Contents

- **Vacuum Technology:**
- Principles of vacuum pumps in range of 10-2torr to 10-11torr, principle of different vacuum pumps, roots pump, rotary, diffusion, turbo molecular pump, cryogenic-pump, ion pump, Ti-sub limitation pump, importance of measurement of vacuum, Concept of different gauges, bayet- albert gauge, pirani, penning, pressure control. [5 Lectures]
- **Conditions for the Formation of Thin Films:**
- Environment for thin film deposition, deposition parameters and their effects on film growth, formation of thin films (sticking coefficient, formation of thermodynamically stable cluster theory of nucleation), capillarity theory, microstructure in thin films, adhesion, properties of thin films, Mechanical, electrical, and optical properties of thin films, few applications of thin films in various fields, Quartz crystal thickness for measurement of film thickness. [5 Lectures]
- **Physical Vapor Deposition Electrical Discharges for Thin Film Deposition:**
- Thermal evaporation, resistive evaporation, Electron beam evaporation, Laser ablation, Flash and Cathodic arc deposition, Sputtering, Glow discharge sputtering, Magnetron sputtering, Ion beam sputtering, Ion plating, oxidizing and Nitriding, Atomic layer deposition (ALD), Importance of ALD technique, Atomic layer growth, Physics and technology. [8 Lectures]
- **Chemical Vapor Deposition Techniques:**
- Advantages and disadvantages of Chemical Vapor deposition (CVD) techniques over PVD techniques, reaction types, boundaries and flow, Different kinds of CVD techniques: Metallorganic CVD (MOCVD), Thermally activated CVD, Spray pyrolysis, etc. [8 Lectures]
- **Lithography and Pattern Transfer:**
- Overview of Lithography, Optics of Lithography: Metrics, Optics of Micro-Lithography, Aligners, Photomasks, Photoresists, Components of Photoresist, Metrics, Photoresist processing, Multi-layer resist, positive, negative and image reversal, Advanced Lithography: E-beam Lithography, Soft Lithography, Etch verses Lift Off. Basic Concepts of Etching, Wet Etching, Specific Wet Etches : Silicon, Silicon Dioxide, Aluminum, Dry (Plasma) Etch etc. [8 Lectures]
- **MEMS and CMOS Manufacturing Technologies:**
- Anisotropic Etching, Description of the Process and Testing, Bulk Micromachining :DRIE-Based, Wet Processes, Surface Micromachining, Wafer Bonding, Introduction of Front end (Fusion Bonding), Back end (Anodic Bonding), Plastic Processes, Molding, Embossing, LIGA, interconnects, CMOS process etc. [8 Lectures]
- **Mini-project:** The students will also carry out a mini-project involving the exposure of lab work. The aim of this project will be to understand, solve and implement solutions to real world problems.

References

1. S. Franssila, **Introduction to Microfabrication**, 2nd Edition, Wiley, 2010.
2. James D. Plummer, Michael D. Deal and Peter B. Griffin, **Silicon VLSI Technology: Fundamentals**, Practice and Modeling, Prentice Hall, 2000.
3. Richard C. Jaeger, **Introduction to Microelectronic Fabrication**, 2nd Edition, Prentice Hall, 2002.
4. Gary S. May, Simon M. Sze, **Fundamentals of Semiconductor Fabrication**, Wiley, 2004.
5. G.L. Weissler and R.W. Carlson, **Methods of Experimental Physics** (Vol 14), Vacuum Physics and Technology.
6. T.A. Delchar, **Vacuum Physics and Techniques**, Chapman and Hall
7. J.P. Hirth and G.M. Pound, **Evaporation: Nucleation and Growth Kinetics, MEMS Manufacturing Technologies**, Pergamon Press